Cisco NetFlow:

- NetFlow is an application for collecting Internet Protocol traffic information.
- NetFlow is a protocol developed by Cisco Systems to record all IP traffic flows.
- In NetFlow, Flow is unidirectional stream of packets from source to destination.
- In Cisco NetFlow, Flow is a stream of packets that share the same characteristics.
- Flow characteristics are like source, destination port, address, protocol, type etc.
- The Cisco NetFlow allows the tracking these flows on in the LAN network.
- NetFlow track the number of packets sent, bytes sent, packet sizes and more.
- Configure router to keep track of all flows and then export them to central server.
- The Cisco NetFlow is a great protocol to get an insight in the network traffic.
- NetFlow allows seeing real time data on who/what is eating the bandwidth.
- NetFlow is used to collect data flows from Cisco routers and switches interfaces.
- NetFlow is application that provides statistics on packets flowing through routers.
- NetFlow captures statistics on Internet Protocol (IP) flows through a device.
- The Cisco NetFlow allows collecting traffic and analyzing it through a program.
- NetFlow is configured in the interface configuration mode on Cisco Routers.
- NetFlow monitor of ingress traffic, egress traffic, or both ingress and egress traffic.
- Specify the IP address of the NetFlow collector & UDP port of collector is listening.
- Provides statistics on packets flowing through Cisco Routers or Cisco Switches.
- NetFlow collect and export the data to enable network and security monitoring.
- NetFlow collect & export data for network planning, traffic analysis & IP accounting.
- NetFlow records are exported to a NetFlow collector using User Datagram Protocol.
- The standard value is the UDP port 2055, but other values can be set 9555 or 9995.
- NetFlow can also, be used for network traffic accounting and for security auditing.
- The Cisco NetFlow consumes additional memory and CPU of devices to process.
- NetFlow is protocol for collecting, aggregating & recording traffic flow data in network.
- NetFlow data provide a more granular view of how bandwidth and network traffic.
- NetFlow was developed by Cisco and is embedded in Cisco's IOS software Routers.
- Many other hardware manufacturers either support NetFlow or use alternative flow.
- There are ten different versions of NetFlow, several versions were released only internally.
- The original NetFlow version 1 is considered obsolete, and seldom used today.
- Versions 2 through 4 were internal versions, no public implementation was ever released.
- Version 5 is still commonly used today, because of a large existing install base device.
- The Cisco NetFlow Version 6 is no longer supported and was not released widely.
- NetFlow version 7 added support for Cisco Catalyst switches using hybrid or native mode.
- NetFlow version 8 has support for when router-based NetFlow aggregation is used.
- The Cisco NetFlow version nine (9) is the current version and is template-based.
- A flow is a way of grouping a unidirectional stream of packets into a specific set.

NetFlow Versions:

Version 1:

- First implementation, now obsolete, and restricted to IPV4 only.

Versions 2:

- NetFlow V2 is Cisco internal version never released to the public.

Version 3:

- NetFlow V3 is Cisco internal version never released to the public.

Version 4:

- NetFlow V4 is Cisco internal version never released to the public.

Version 6:

- NetFlow V 6 is no longer supported by Cisco Encapsulation information.
- NetFlow Version 6 is not compatible with Cisco Routers /Switches obsolete.

Version 7:

- NetFlow V7 is Cisco-specific version for Catalyst 5000 series Switches.
- NetFlow Version 7 is not compatible with Cisco Routers obsolete.

Version 8:

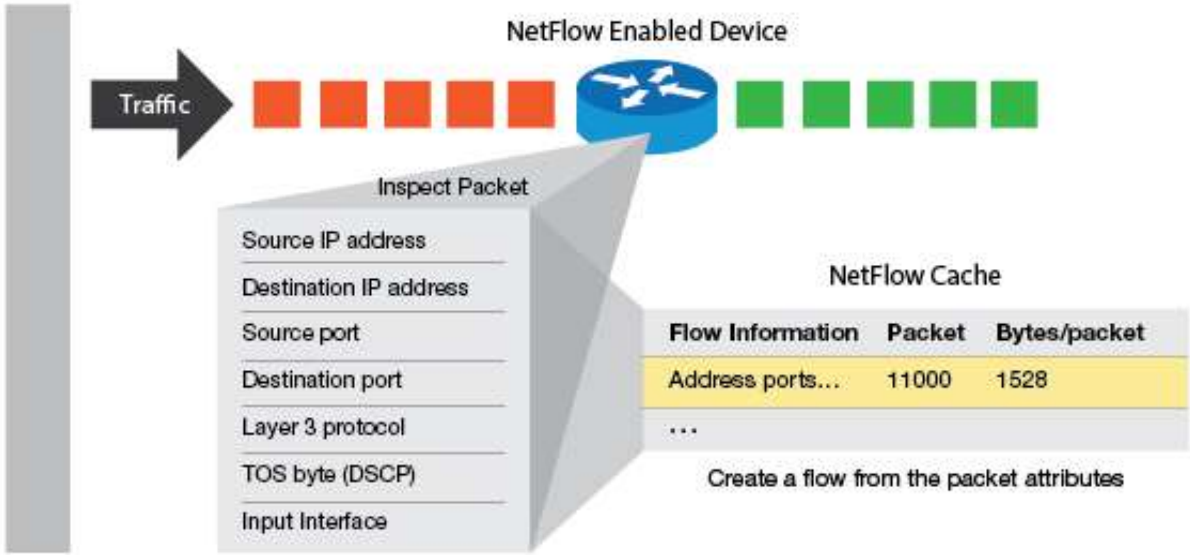
- Choice of aggregation schemes in order to reduce resource usage.
- NetFlow V8 is no longer supported by Cisco it is totally obsolete.

Version 5:

- Fixed format that cannot be added or extended.
- NetFlow version 5 only support Inter Protocol v4.
- NetFlow version 5 added the BGP support as well.
- NetFlow Version 5 export data from main cache only.
- NetFlow V 5 no real concept of ingress and egress flows.
- NetFlow added flow sequence numbers & additional fields.
- NetFlow Version 5 is standard & most common NetFlow version.

Version 9:

- NetFlow Version 9 support IPV4 and IPv6.
- Not backwards compatible with any previous version.
- Added additional information to flows & template based.
- Exports data from main & aggregation cache.
- NetFlow Version 9 support for MPLS.
- Support flow-record format known as Flexible NetFlow technology.
- NetFlow Version 9 is most important NetFlow version.



NetFlow uses the following fields to identify a unique flow:

- Source IP Address
- Source Port Number
- Destination IP Address
- Destination Port Number
- Layer 3 Protocol Type
- Type of Service
- Logical Input Interface

Flow:

- A flow is a way of grouping a unidirectional stream of packets into a specific set.
- As each packet is forwarded, the above-mentioned attributes are examined.
- A flow is generated by the first packet passing through the standard switching path.
- Each additional packet with the same parameters is grouped into a single flow.
- Any variation in the value of any one of the parameters creates a new flow.
- Routers support sampled NetFlow only one out of certain number of packets is examined.
- This is for use on routers examining every packet is impractical due to volume of traffic.
- Sampled flows significantly reduce the performance impact when sending flow information.

NetFlow Cache:

- Monitoring & grouping every packet forwarded by router generates a lot of data.
- This data is condensed into a database within network device called the NetFlow cache.
- Record is kept for each active flow; data is expired & then exported to NetFlow collector.

NetFlow Uses:

The most obvious use for NetFlow is network monitoring. NetFlow data provides detailed bandwidth usage information that can be segmented in numerous ways, including by user, client system, time and application. The data arriving at the NetFlow collector is near-real time, allowing for specific granular monitoring and for aggregating data to look at the big picture as it is happening. Monitoring traffic patterns, user patterns and application patterns can alert an administrator to potential problems before they happen & provide a valuable troubleshooting resource. A single computer or service using a sufficiently large amount of bandwidth can affect network performance for other users.

- Network monitoring
- User and application monitoring
- Application and user profiling
- Capacity planning
- Identifying security anomalies
- Network data mining
- Network planning
- Network fault troubleshooting
- Usage-based billing and reporting
- Application reporting and profiling
- Security analysis
- Enhanced Network Visibility
- Enabling Root Cause Diagnosis
- Security Awareness